

Lecture 23

ChE 467/596-012 **Polymer Rheology**

Saad A Khan

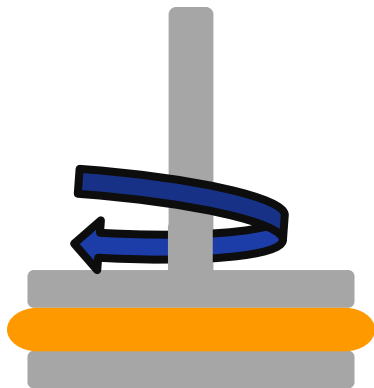
Today's Agenda

Review via in class problems

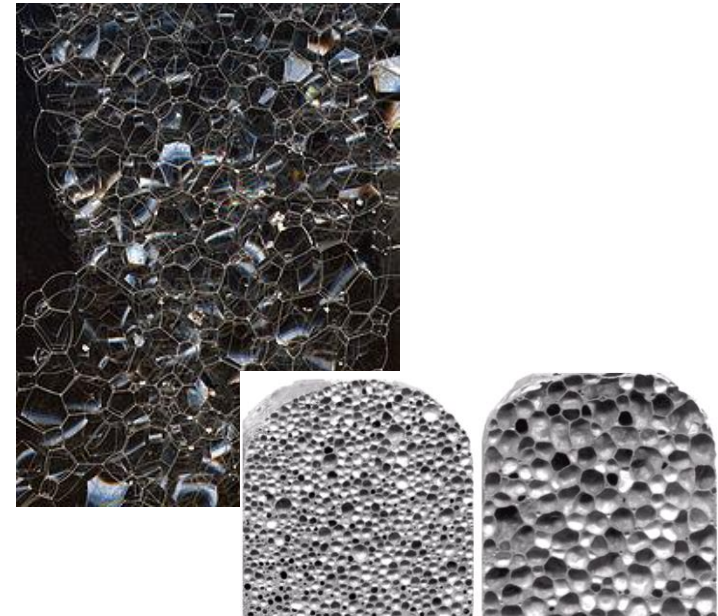
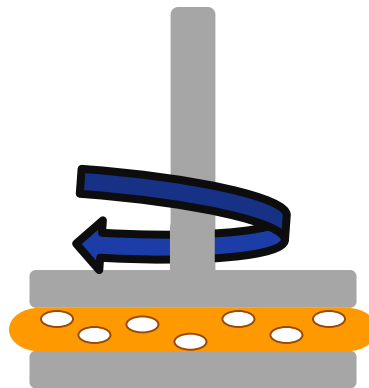
Rheology of Foams

What is a foam?

Substance that is made by trapping air pockets within a liquid or solid



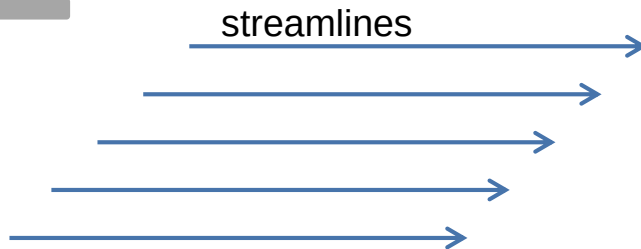
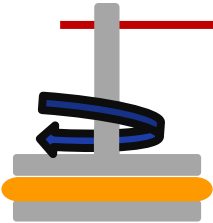
VS.



How does the viscosity of a liquid change when air is added to it?

Rheology of Foams

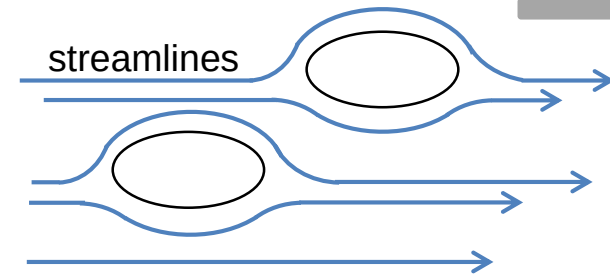
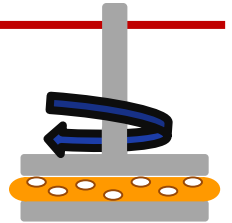
Liquid Solution



Streamlines of flow are straight

Energy from stress is used to strain the material

Foam



Streamlines of flow move around air bubbles in solution

Energy dissipates when maneuvering around bubbles

Amount of strain from a given stress is reduced

Viscosity _____ with bubble incorporation

A) Decreases

B) Remains the same

C) Increases

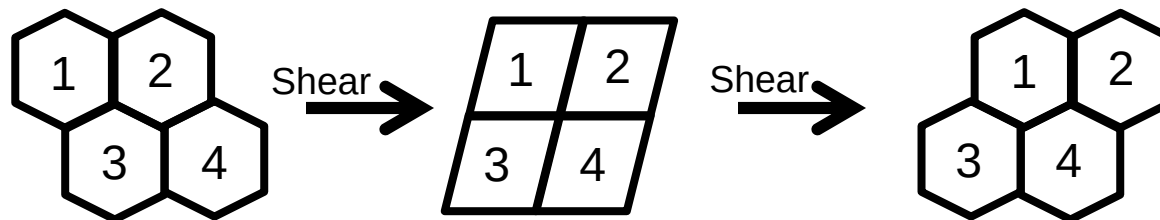
D) Goes crazy

Rheology of Foams: Problem

Foams of the same material and bubble/cell sizes were prepared with different volume fractions, ϕ_A and ϕ_B , where $\phi_A > \phi_B$. How will the viscosities of the samples compare? What happens to the viscosity as the volume fraction increases even more?

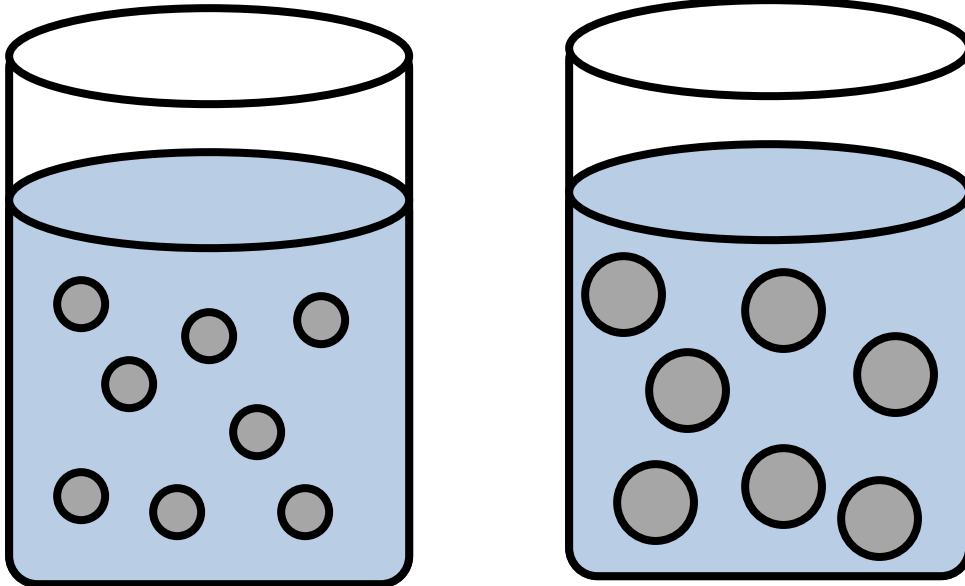


As volume fraction increases ($\phi \geq 0.64$), foams exhibit yield stress behavior due to close packing of bubbles and cells.



Dilute Suspensions: Problem

Two suspensions, both at $\phi = 0.001$, were prepared using glass beads. One had beads of diameter 2 micron and the other had beads of diameter 3 micron. Peter thinks that the sample containing the beads of 3 micron diameter will have a higher viscosity than the sample with 2 micron diameter beads. Is he correct? Why or why not?



$\phi = 0.001 \rightarrow$ dilute solution

Apply Einstein's equation:

$$\eta = \eta_s \left[1 + \frac{5}{2} \phi \right]$$

$\rightarrow$ Viscosity is independent of bead size

Einstein's Equation

$$\eta = \eta_s \left[1 + \frac{5}{2} \phi \right]$$

Assumptions:

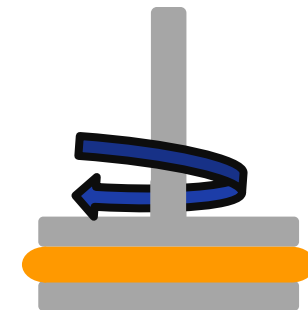
1. Creeping flow
2. Surrounding fluid is Newtonian
3. No particle settling
4. No particle migration
5. No wall effects
6. Rigid spheres
7. No slip between particle and fluid

Einstein's Equation

$$\eta = \eta_s \left[1 + \frac{5}{2} \phi \right]$$

Assumption 1: Creeping flow ($Re \ll 0.1$)

$$\text{Reynolds Number} = \frac{\text{inertial forces}}{\text{viscous forces}}$$



At low Reynolds number, inertial effects negligible

At high Reynolds number,

$$\eta = \eta_s \left[1 + \frac{5}{2} \phi + 1.34 Re_p^{3/2} \right]$$

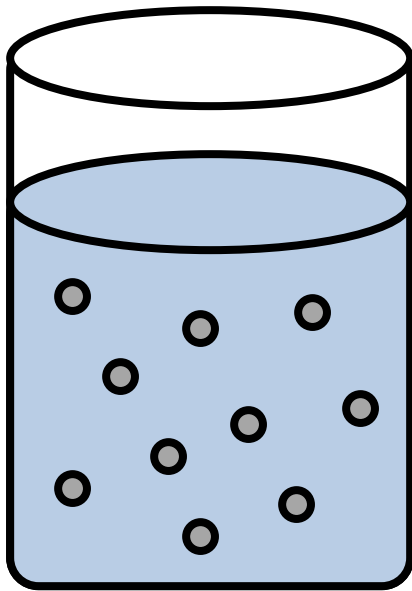
$$\text{where } Re_p = \frac{\rho \dot{\gamma} a^2}{\eta_s}$$

Assumption 5: No wall effects

Dilute Suspensions: Problem 2

If the beads had a diameter of 0.5 micron, how would the viscosity compare to the previous samples?

Deviation from Einstein's equation as
sphere size $\leq 1 \mu\text{m}$



WHY?

Einstein's equation only applies to suspensions where particles are not interacting

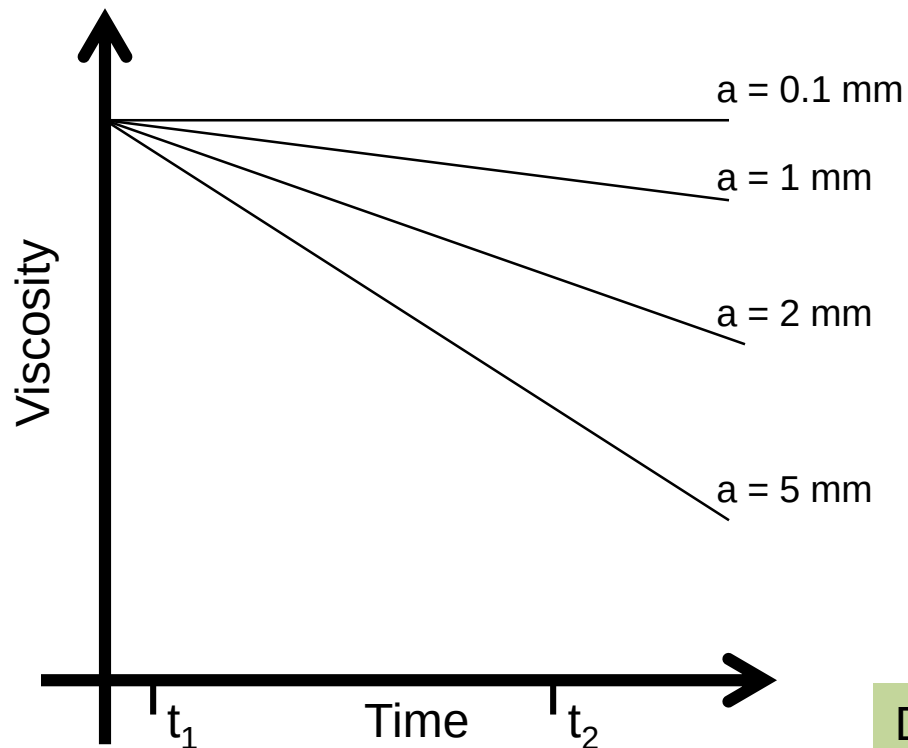
Must take interactions between particles into consideration:

Use Generalized Einstein Equation:

$$\eta = \eta_s \left[1 + \frac{5}{2} \phi + k \phi^2 + \dots \right]$$

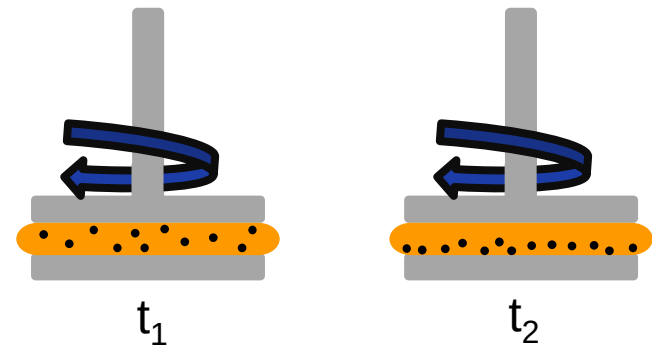
Dilute Solutions: Problem 3

The zero shear viscosity of a dilute suspension of solid particles ($\phi = 0.05$) is measured at different time intervals using couette geometry. The following figure shows variations in the zero shear viscosity of the suspension for different particle sizes.



(a) Explain why the zero shear viscosity decreases with time.

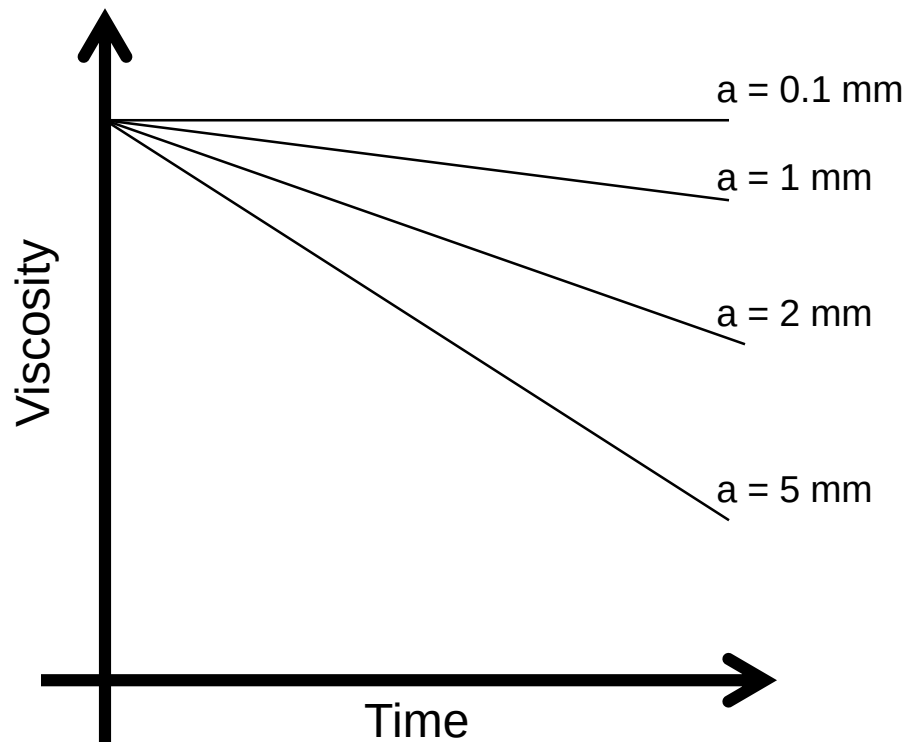
Particles are settling out of the solution over time.



Desire time for settle $\gg$ time for experiment

Dilute Solutions: Problem 3

The zero shear viscosity of a dilute suspension of solid particles ($\phi = 0.05$) is measured at different time intervals using couette geometry. The following figure shows variations in the zero shear viscosity of the suspension for different particle sizes.

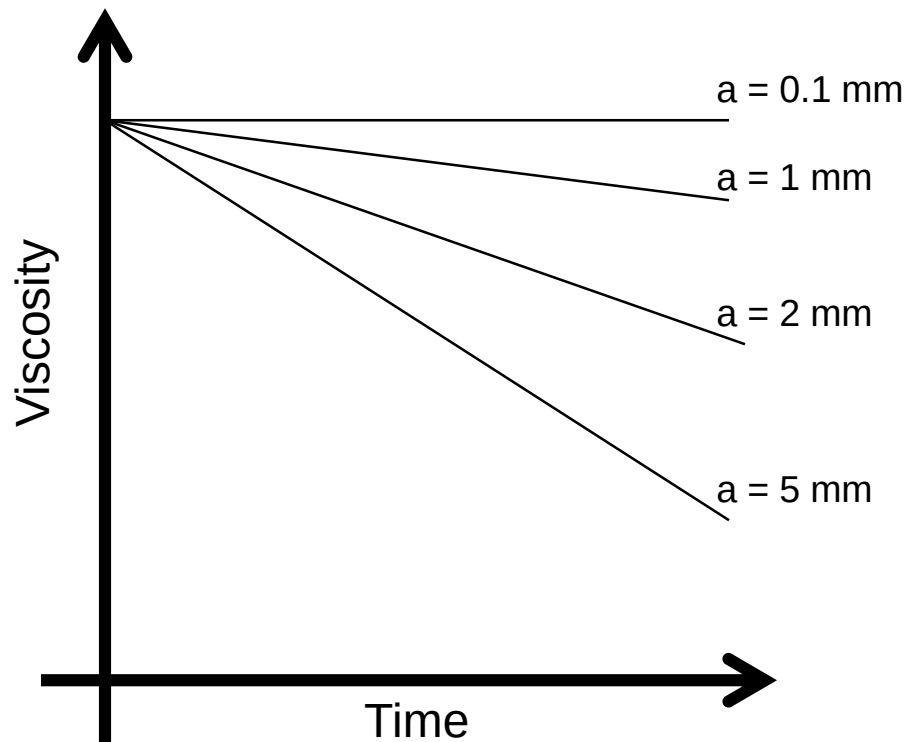


(b) Explain why the zero shear viscosity does not seem to decrease over time for particles of size 0.1 mm.

Brownian motion and interactions occur between smaller particles. Prevents particles from settling over time of experiment.

Dilute Solutions: Problem 3

The zero shear viscosity of a dilute suspension of solid particles ($\phi = 0.05$) is measured at different time intervals using couette geometry. The following figure shows variations in the zero shear viscosity of the suspension for different particle sizes.



(c) Explain why the rate of decrease in the viscosity increases with particle size.

$$t_{0.1h} = \frac{0.45\eta_s h}{\left|(\rho_p - \rho_s)\right| a^2 g}$$

a = particle radius

h = gap or height (Couette)

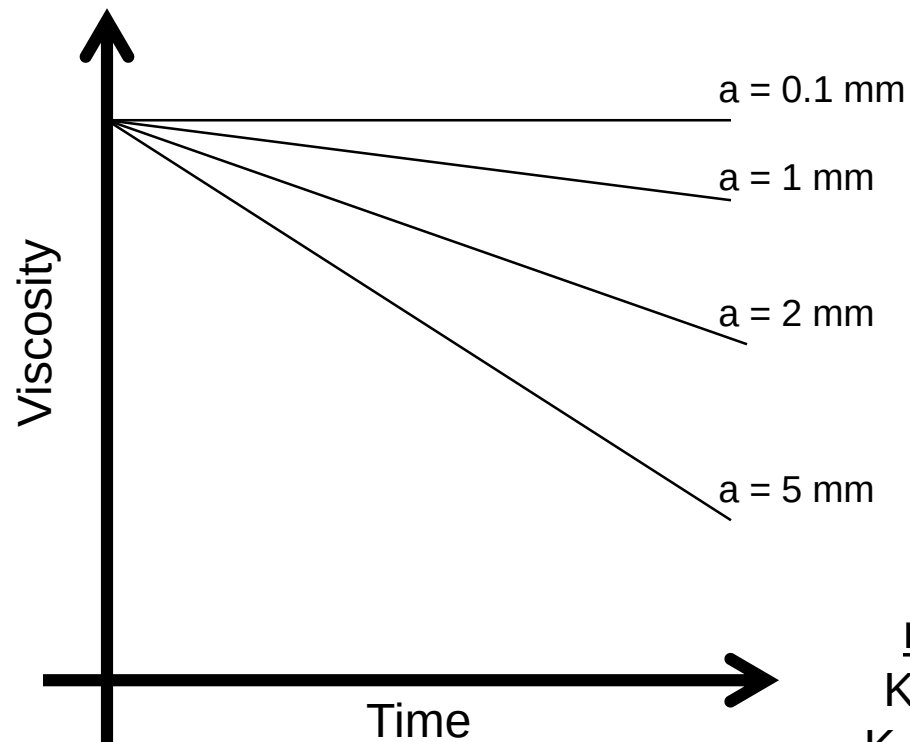
η_s = solvent viscosity

ρ_p = density of particle

ρ_s = density of solvent

Dilute Solutions: Problem 3

The zero shear viscosity of a dilute suspension of solid particles ($\phi = 0.05$) is measured at different time intervals using couette geometry. The following figure shows variations in the zero shear viscosity of the suspension for different particle sizes.



(c) Explain why the rate of decrease in the viscosity increases with particle size.

$$K = \frac{\rho_p \bar{v}^2 a^4}{hkT}$$

ρ_p = density of particle

a = particle radius

$v = \frac{1}{2} v_{\max}$ (Couette)

$v = \frac{2}{3} v_{\max}$ (Poiselle)

h = gap or height (Couette)

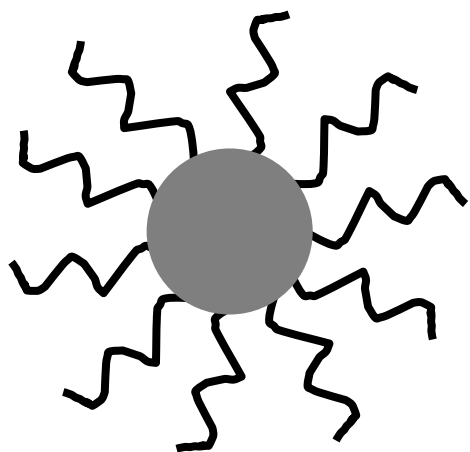
k = Boltzmann constant

T = Temperature

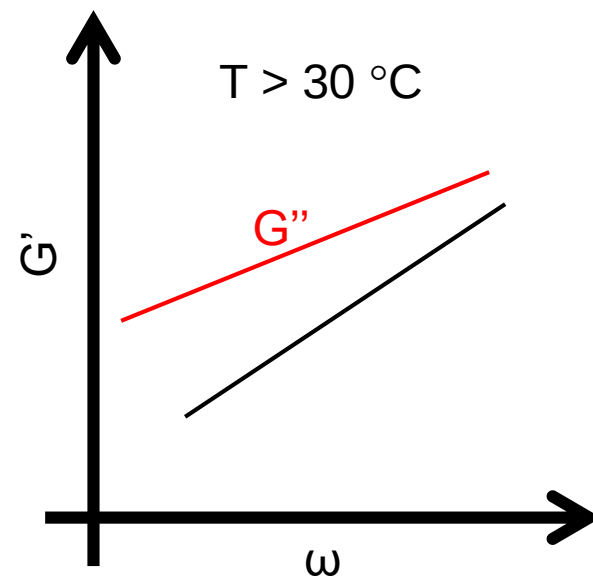
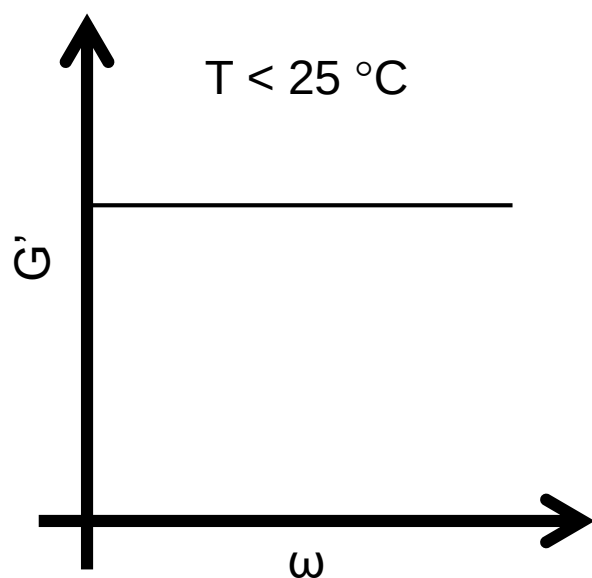
Neglect migration if:
 $K < 0.1$ (Couette)
 $K < 0.01$ (Poiselle)

Suspension and Flocculation: Problem

Silica (SiO_2) particles bearing octadecyl (C_{18}) chains on the surface are suspended in hexadecane. The rheology of the system is highly dependent on temperature. At 30°C , the system shows solution-like behavior. At 25°C and below, the system is gel-like.

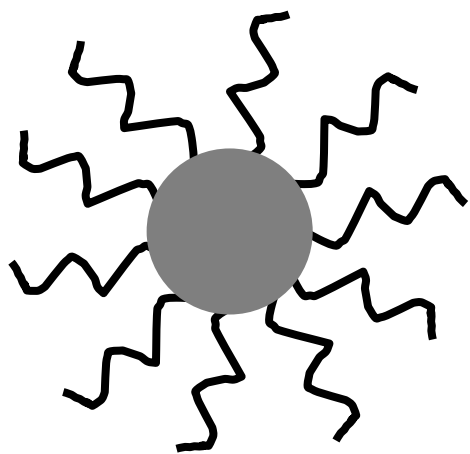


(a) What conclusions can we make about the elastic modulus dependence on frequency at temperatures below 25°C and above 30°C ?



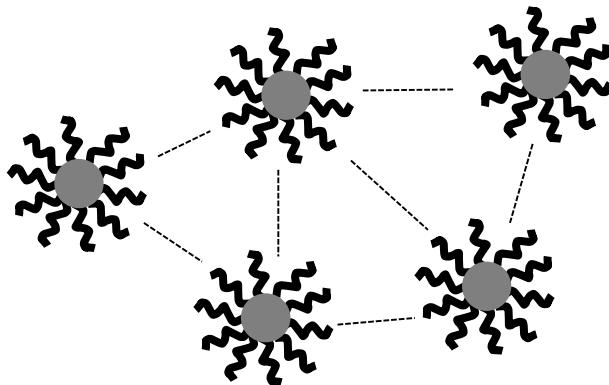
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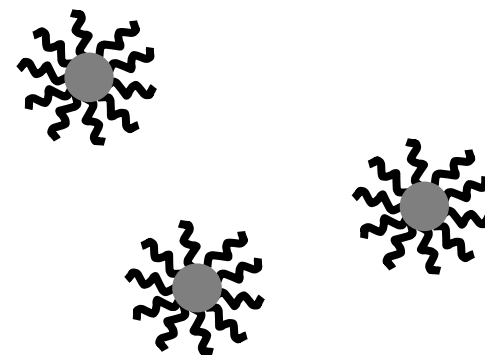
(b) What conclusions can we make about the interactions between particles at temperatures below 25 °C and above 30 °C?

$T < 25\text{ °C}$



Attractive forces (van der Waals)
exceed Brownian forces

$T > 30\text{ °C}$



Particles sterically
stabilized

Foam Problem

Draining of Foam (Sp 2015)

Stephanie from the Velev group has prepared a monodisperse foam at a gas volume fraction of 0.25. Please help her out by validating some issues.

- (a) What would the viscosity versus shear rate profile look like? Please show a qualitative sketch.
- (b) After many experiments, Stephanie has ran out of solvent. She decides to switch to a new solvent of approximately the same viscosity to make a fresh batch of foams (the volume fraction is the same as before). How will this affect the viscosity vs. shear rate plot? Why?

Foam Problem

Foam at a gas volume fraction of 0.25.

- (a) What would the viscosity versus shear rate profile look like? Please show a qualitative sketch.

Foam Problem

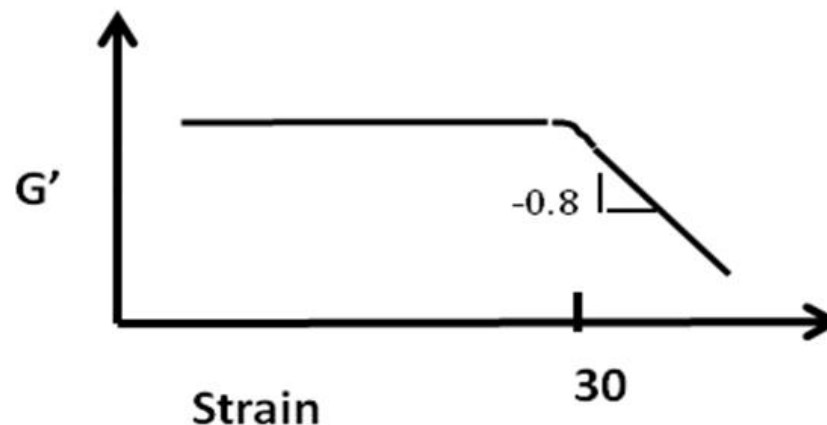
Foam at a gas volume fraction of 0.25.

(b) After many experiments, Stephanie has ran out of solvent. She decides to switch to a new solvent of approximately the same viscosity to make a fresh batch of foams (the volume fraction is the same as before). How will this affect the viscosity vs. shear rate plot? Why?

Suspension Problem

Mr. TN has added some submicron fillers to liquid metal (think of liquid metal as a solvent) at the suggestion of Mr. DP. He has obtained the following data in terms of G' versus strain, running the experiment at a constant frequency of 0.1 rad/s.

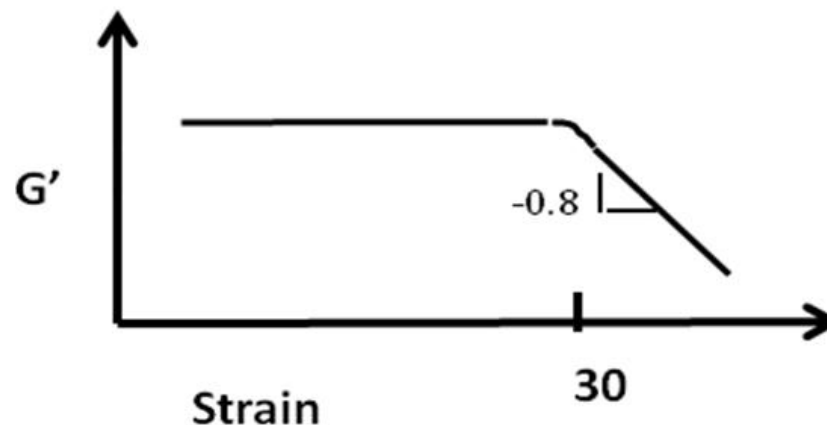
(a) Both Mr. TN and Mr. DP are very intrigued by the data. They think that the sample is a gel. Please enlighten them with your thoughts, arguments and wisdom. Do you think it is a gel, why, why not? Are there other experiments they should run, or are they good?



Suspension Problem

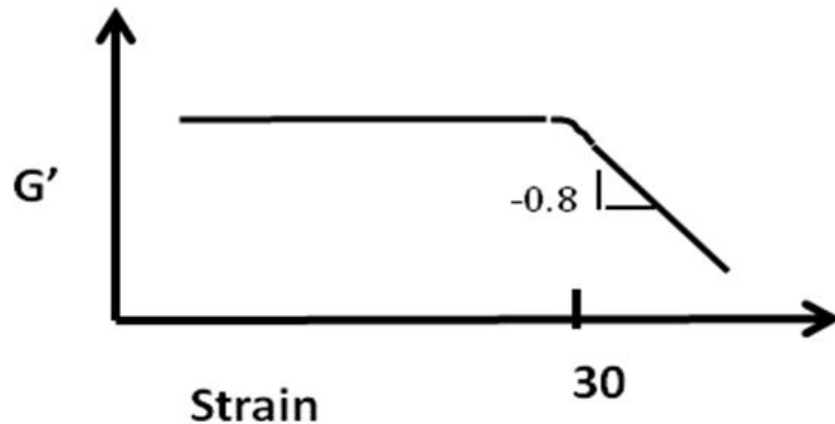
Mr. TN has added some submicron fillers to liquid metal (think of liquid metal as a solvent) at the suggestion of Mr. DP. He has obtained the following data in terms of G' versus strain, running the experiment at a constant frequency of 0.1 rad/s.

(b) Mr. TN and Mr. DP also think that this material has a yield stress, and they can back out from the data the yield stress as well as the strain at which the sample yields. Please explain if these gentlemen are correct, why and why not? If it has a yield stress, how can it be determined from this data, as well as how they can determine the yield strain?



Suspension Problem

(b) Mr. TN and Mr. DP also think that this material has a yield stress, and they can back out from the data the yield stress as well as the strain at which the sample yields. Please explain if these gentlemen are correct, why and why not? If it has a yield stress, how can it be determined from this data, as well as how they can determine the yield strain?



Colloids Problem

Colloidal Suspension: Yunhu from the Hsiao group has been working with an unknown colloidal particle (named BQ) supplied to him from a company. Since very little is known about BQ except size, 0.8 microns in diameter, Yunhu is trying to decipher its rheological behavior in various solvents (e.g., acting as a hard sphere, have electrostatic, van der Waals, static interactions, etc).

(a) (10) A high school intern working for Yunhu mixes 0.15 volume fraction of BQ in a Newtonian solvent of viscosity 4 Pa.s. He then measures the viscosity and says that he found the suspension to have a viscosity of 5.5 Pa.s. However, when Yunhu measures the same sample, he gets a viscosity of 4.75 Pa.s. Are any of these values possible or both making mistakes in their measurements? If one is correct, who is correct, and if one of these measurements is accurate, can you say what is happening in terms of rheological behavior. Explain your answer.

Colloids Problem

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(a) 0.15 volume fraction of BQ in a Newtonian solvent of viscosity 4 Pa.s. Intern finds suspension viscosity to be 5.5 Pa.s. Yunhu gets a viscosity of 4.75 Pa.s. What is going on?

Colloids Problem

Colloidal Suspension: Yunhu from the Hsiao group has been working with an unknown colloidal particle (named BQ) supplied to him from a company. Since very little is known about BQ except size, 0.8 microns in diameter, Yunhu is trying to decipher its rheological behavior in various solvents (e.g., acting as a hard sphere, have electrostatic, van der Waals, static interactions, etc).

(b) (10) Dr. Hsiao walks into the lab with two more samples BQ1 and BQ2. She says that one has charges on its surface while the other does not; however, she does not which one is which. Yunhu puts 0.15 volume fraction of each colloid in water. He finds the viscosity of BQ1 > BQ2. However, when he puts 0.15 volume fraction of each colloid in nonpolar oil, he finds the viscosity of BQ1 and BQ2 to be essentially same. Please explain what is going on in each case; is there any way you can determine from these experiments which colloid has electrostatic charges. Please justify your answer. [Note that the experimental results are accurate, i.e., there were no errors in the measurements; so this is not an answer]

Colloids Problem

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(b) Two more samples BQ1 and BQ2, one has charges on its surface. Yunhu puts 0.15 volume fraction of each colloid in water. He finds the viscosity of BQ1 > BQ2. However, when he puts 0.15 volume fraction of each colloid in nonpolar oil, he finds the viscosity of BQ1 and BQ2 to be essentially same. Please explain.